

Baseline Simulation DGP and Assumption Audit

The Law of Portfolio Learnability

Implementation specification for Codex

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Purpose. This note defines one theorem-compatible baseline simulation economy. The DGP is chosen first; the population portfolio rule is then derived from the portfolio loss. No form for W^* is imposed ex ante. The construction gives Gaussian returns, a finance-style rank normalization, an exact conditional pricing restriction, and a Matérn managed-payoff spectrum.

1 Fixed dimensions and kernel

Use

$$D = 6, \quad N = 3G, \quad K_F = 3.$$

The characteristic space is $\mathcal{Z} = (-1, 1)^D$. Use a Matérn-3/2 kernel

$$K(z, z') = \left(1 + \frac{\sqrt{3} \|z - z'\|}{\ell}\right) \exp\left(-\frac{\sqrt{3} \|z - z'\|}{\ell}\right), \quad K(z, z) = 1,$$

with fixed length scale $\ell > 0$. Hence $\kappa_K = 1$. For $D = 6$ and $\nu = 3/2$, the corresponding population Matérn integral operator has eigenvalue exponent

$$b = 1 + \frac{2\nu}{D} = \frac{3}{2}.$$

2 Persistent characteristics and rank normalization

For groups $g = 1, \dots, G$, coordinates $d = 1, \dots, D$, generate independent stationary Gaussian AR(1) states

$$S_{g,d,t+1} = \rho_d S_{g,d,t} + \sqrt{1 - \rho_d^2} \eta_{g,d,t+1}, \quad \eta_{g,d,t+1} \stackrel{iid}{\sim} N(0, 1), \quad |\rho_d| < 1,$$

with $S_{g,d,0} \sim N(0, 1)$. Convert them to population ranks

$$U_{g,d,t} = \Phi(S_{g,d,t}) \in (0, 1).$$

Each group generates three stocks, indexed by $r \in \{0, 1, 2\}$. For stock $i = (g, r)$ define the phase-balanced percentile

$$U_{i,d,t} = (U_{g,d,t} + r/3) \bmod 1, \quad Z_{i,d,t} = 2U_{i,d,t} - 1.$$

Thus every stock has a continuous uniform marginal characteristic on $(-1, 1)^D$, while the three stocks in each group are deliberately cross-sectionally dependent. This is the theorem-compatible rank construction. A literal empirical cross-sectional rank transform may be used only as a robustness simulation, not for the exact spectral audit, because for fixed N it produces a finite support.

3 Balanced smooth factor loadings

Let $u(z) = (z_1 + 1)/2$ and fix $c_\beta > 0$. Define

$$\beta(z) = \sqrt{c_\beta} \begin{pmatrix} \sqrt{2/3} \cos(2\pi u(z)) \\ \sqrt{2/3} \sin(2\pi u(z)) \\ 1/\sqrt{3} \end{pmatrix} \in \mathbb{R}^3.$$

Then, for every z ,

$$\|\beta(z)\|^2 = c_\beta.$$

Let $B_t \in \mathbb{R}^{N \times 3}$ have row i equal to $\beta(Z_{i,t})'$. Because the three phase shifts are 0, 1/3, 2/3, every group satisfies

$$\sum_{r=0}^2 \beta(Z_{g,r,t}) \beta(Z_{g,r,t})' = c_\beta I_3.$$

Therefore, exactly for every date,

$$\boxed{\frac{1}{N} B_t' B_t = \Gamma_\beta := \frac{c_\beta}{3} I_3.}$$

This exact balance is the key device that makes the conditional optimizer representable by one time-invariant characteristic policy.

4 Factors, idiosyncratic shocks, and returns

Fix $\mu_F \in \mathbb{R}^3$ with $0 < \|\mu_F\| < 1$. Generate

$$F_{t+1} \stackrel{iid}{\sim} N(\mu_F, I_3 - \mu_F \mu_F'),$$

independently of the characteristic process. Hence

$$\mathbb{E}[F_{t+1}] = \mu_F, \quad \mathbb{E}[F_{t+1} F_{t+1}'] = I_3.$$

Generate idiosyncratic shocks

$$\varepsilon_{t+1} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2 I_N)$$

independently of everything else. Returns are

$$\boxed{R_{t+1} = B_t F_{t+1} + \varepsilon_{t+1}.}$$

Let $\mathcal{F}_t$ contain the complete history through date t and the current Z_t , but not F_{t+1} or ε_{t+1} . Then

$$R_{t+1} \mid \mathcal{F}_t \sim N(m_t, V_t),$$

where

$$m_t = B_t \mu_F, \quad V_t = B_t (I_3 - \mu_F \mu_F') B_t' + \sigma_\varepsilon^2 I_N,$$

and the conditional second moment is

$$\boxed{S_t := \mathbb{E}[R_{t+1} R_{t+1}' \mid \mathcal{F}_t] = B_t B_t' + \sigma_\varepsilon^2 I_N.}$$

5 Population portfolio rule derived from the loss

For a characteristic rule $W : \mathcal{Z} \rightarrow \mathbb{R}$, define

$$w_t(W) = \frac{1}{N} (W(Z_{1,t}), \dots, W(Z_{N,t}))', \quad R_{t+1}^p(W) = w_t(W)' R_{t+1}.$$

The conditional quadratic loss is

$$Q_t(W) = \mathbb{E}[(1 - R_{t+1}^p(W))^2 \mid \mathcal{F}_t] = 1 - 2w_t(W)' m_t + w_t(W)' S_t w_t(W).$$

The unconstrained conditional minimizer satisfies $S_t w_t^* = m_t$. Using

$$(B_t B_t' + \sigma_\varepsilon^2 I_N)^{-1} B_t = B_t (B_t' B_t + \sigma_\varepsilon^2 I_3)^{-1}$$

and $B_t' B_t / N = \Gamma_\beta$,

$$w_t^* = \frac{1}{N} B_t \left(\Gamma_\beta + \frac{\sigma_\varepsilon^2}{N} I_3 \right)^{-1} \mu_F.$$

Hence the minimizer belongs to the characteristic-policy class and is generated by the single population function

$$\boxed{W^*(z) = \beta(z)' \left(\Gamma_\beta + \frac{\sigma_\varepsilon^2}{N} I_3 \right)^{-1} \mu_F = \frac{\beta(z)' \mu_F}{c_\beta/3 + \sigma_\varepsilon^2/N}.}$$

This formula is derived from the return DGP and the portfolio loss; it is not imposed.

6 Audit of Conditions E1–E6

E1: stationarity and mixing. The finite-dimensional latent Gaussian AR(1) state is geometrically β -mixing because all $|\rho_d| < 1$. The rank/phase map is measurable, so it cannot increase absolute-regularity coefficients. Adding independent one-step Gaussian factor and idiosyncratic shocks preserves geometric mixing. Therefore (Z_t, R_{t+1}) is geometrically β -mixing and hence satisfies $\beta(k) \leq C_\gamma k^{-\gamma}$ for every finite γ . In particular choose any $\gamma > b = 3/2$.

E2(a): Gaussian-type managed-feature tails. Let $K_t = (K(Z_{i,t}, Z_{j,t}))_{i,j}$ and

$$X_t = \frac{1}{N} \sum_{i=1}^N R_{i,t+1} K(\cdot, Z_{i,t}), \quad \|X_t\|_{\mathcal{H}_K}^2 = \frac{1}{N^2} R'_{t+1} K_t R_{t+1}.$$

Because $K_t \succeq 0$ and $K(z, z) \leq \kappa_K$,

$$\|K_t\|_{\text{op}} \leq \text{tr } K_t \leq N \kappa_K.$$

Conditionally on $\mathcal{F}_t$, R_{t+1} is Gaussian. Put $A_t = K_t/N^2$. Then

$$\|A_t\|_{\text{op}} \leq \frac{\kappa_K}{N}, \quad \mathbb{E}[\|R_{t+1}\|^2 \mid \mathcal{F}_t] = N(c_\beta + \sigma_\varepsilon^2),$$

and

$$\|V_t\|_{\text{op}} \leq N c_\beta / 3 + \sigma_\varepsilon^2.$$

For a Gaussian quadratic form, the standard mgf formula and the inequalities $-\log(1-x) \leq 2x$ for $0 \leq x \leq 1/2$ give the uniform sufficient choice

$$\boxed{B_X^2 = 4\kappa_K(c_\beta + \sigma_\varepsilon^2)}$$

for which

$$\boxed{\mathbb{E} \left[\exp \left(\frac{\|X_t\|_{\mathcal{H}_K}^2}{B_X^2} \right) \middle| \mathcal{F}_t \right] \leq e^{1/2} < 2.}$$

Thus the paper's Gaussian-type tail condition holds with unbounded Gaussian returns.

E2(b): conditional fourth moments. For every $V \in \mathcal{H}_K$, $R_{t+1}^p(V) \mid \mathcal{F}_t$ is scalar Gaussian. If $Y \sim N(a, s^2)$, then

$$\mathbb{E}[Y^4] = a^4 + 6a^2s^2 + 3s^4 \leq 3(a^2 + s^2)^2 = 3\{\mathbb{E}[Y^2]\}^2.$$

Therefore the fourth-moment condition holds with

$$\boxed{K_4 = 3.}$$

E3: RKHS membership of the population policy. Each component of β is smooth (trigonometric or constant) on the compact cube. For a Matérn- ν kernel on a bounded regular domain, the RKHS is equivalent to a Sobolev space of order $s = \nu + D/2$. With $\nu = 3/2$ and $D = 6$, $s = 9/2$. Hence $\beta_k \in \mathcal{H}_K$ for $k = 1, 2, 3$, and because W^* is a fixed linear combination of them,

$$\boxed{W^* \in \mathcal{H}_K, \quad \|W^*\|_{\mathcal{H}_K} < \infty.}$$

For a class of DGPs, keep $\|\mu_F\|$, c_β , σ_ε and N in compact ranges bounded away from degeneracy to obtain a common R .

E4: conditional pricing restriction. By construction, $S_t w_t^* = m_t$. Hence for every $V \in \mathcal{H}_K$,

$$\mathbb{E}[(1 - R_{t+1}^p(W^*)) R_{t+1}^p(V) \mid \mathcal{F}_t] = w_t(V)'(m_t - S_t w_t^*) = 0.$$

Thus E4 holds exactly, not approximately.

E5: polynomial managed-payoff spectrum. Let

$$T_K := \mathbb{E}[K_Z \otimes K_Z], \quad \Sigma_{\mathcal{H}_K} := \mathbb{E}[X_t \otimes X_t].$$

The idiosyncratic component gives exactly $(\sigma_\varepsilon^2/N)T_K$. The factor component is positive and, for every $h \in \mathcal{H}_K$,

$$\mathbb{E} \left\| \frac{1}{N} \sum_i \beta(Z_{i,t}) h(Z_{i,t}) \right\|^2 \leq \mathbb{E} \left[\frac{1}{N} \sum_i \|\beta(Z_{i,t})\|^2 h(Z_{i,t})^2 \right] = c_\beta \langle h, T_K h \rangle.$$

Therefore

$$\boxed{\frac{\sigma_\varepsilon^2}{N} T_K \preceq \Sigma_{\mathcal{H}_K} \preceq \left(c_\beta + \frac{\sigma_\varepsilon^2}{N} \right) T_K.}$$

By the min-max principle, the eigenvalues of $\Sigma_{\mathcal{H}_K}$ have the same decay exponent as those of T_K . For Matérn-3/2 on a six-dimensional compact domain with a density bounded above and below,

$$\lambda_j(T_K) \asymp j^{-3/2},$$

so

$$\boxed{\mu_j(\Sigma_{\mathcal{H}_K}) \asymp j^{-3/2}; \quad b = 3/2.}$$

E6: nondegenerate optimal Sharpe ratio. Let $g = c_\beta/3$. Since $\Gamma_\beta = gI_3$,

$$q^* := \mathbb{E}[R_{t+1}^p(W^*) \mid \mathcal{F}_t] = \mathbb{E}[(R_{t+1}^p(W^*))^2 \mid \mathcal{F}_t] = \frac{g}{g + \sigma_\varepsilon^2/N} \|\mu_F\|^2.$$

This is constant over time. Since $0 < \|\mu_F\| < 1$, we have $0 < q^* < 1$, and

$$\boxed{(SR^*)^2 = \frac{q^*}{1 - q^*}.}$$

For a uniform class assumption, restrict $\|\mu_F\|$ to $[\underline{m}, \bar{m}]$ with $0 < \underline{m} < \bar{m} < 1$.

7 Implementation invariants and required tests

Codex must test the following identities before running Monte Carlo experiments:

1. For every simulated date, $\max_i \|\beta(Z_{i,t})\|^2 - c_\beta < 10^{-12}$.
2. For every simulated date, $\|B'_t B_t / N - (c_\beta/3) I_3\|_{\text{op}} < 10^{-12}$.
3. Numerically verify $\|S_t w_t^* - m_t\| / \max(1, \|m_t\|) < 10^{-11}$.
4. Verify the explicit W^* reproduces every coordinate of w_t^* to machine precision.
5. Check $I_3 - \mu_F \mu_F' \succ 0$ before simulation.
6. Compute q^* and SR^* analytically and compare to a long-run Monte Carlo estimate.
7. Numerically audit E2(a) from the exact Gaussian quadratic-form log-mgf for random dates; do not rely only on Monte Carlo tail counts.
8. Numerically verify the fourth-to-second moment ratio of managed Gaussian payoffs never exceeds 3 up to numerical tolerance.
9. Estimate the leading eigenvalues of the managed operator and of T_K using the same Nyström discretization; confirm the eigenvalue-ratio sandwich and report log-log slopes over a predeclared interior range. The theoretical target is $-3/2$.

8 Suggested baseline calibration

A finance-like monthly baseline may use

$$N = 600, \quad D = 6, \quad \rho_d = 0.95, \quad \nu = 3/2, \quad \ell = 1, \\ c_\beta = (0.04)^2, \quad \sigma_\varepsilon = 0.08, \quad \mu_F = (0.10, 0.06, 0.03)'.$$

These numbers are calibration choices, not theoretical restrictions. With them, $\|\mu_F\| < 1$ and the exact population Sharpe ratio is available from the formula above. Robustness may vary ρ_d , ℓ , c_β , σ_ε , N , and μ_F without changing the DGP logic.

9 Do not change

Do not redesign the paper's loss, estimator, Sharpe definition, theorem statements, or RKHS normalization. Do not impose a population W^* by hand. Do not reintroduce thousands of economic factors or prescribe the managed eigenvalues directly. The only three economic factors are the coordinates of F_{t+1} ; the Matérn spectrum must emerge from the kernel and the characteristic distribution.